

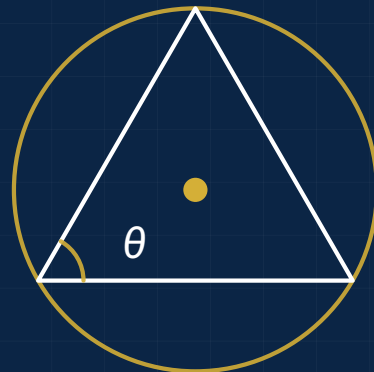


DR ESLAM AHMED
eliteigcse.com

CHAPTER 4 • GEOMETRY & MEASURES • TOPIC 5

Area & Perimeter

Split every shape into rectangles, triangles, trapeziums and circle parts. Area adds the pieces; perimeter walks only round the outside.



4

VARIANTS

22

PAST-PAPER PARTS

46

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Standard shapes

V2 Compound shapes

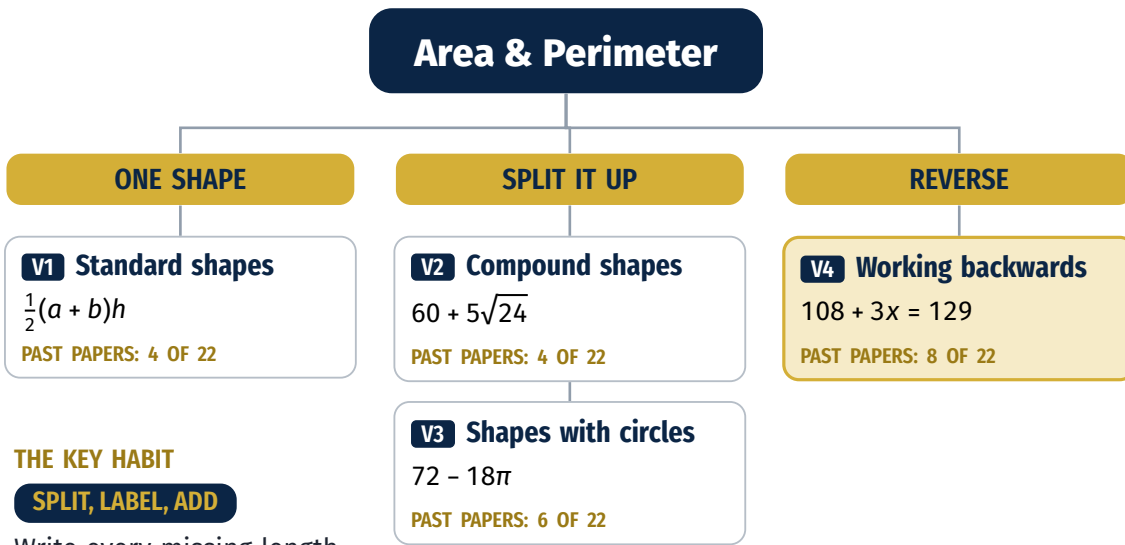
V3 Shapes with circles

V4 Working backwards

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Area & perimeter at a glance



Your notes
**THE KEY HABIT****SPLIT, LABEL, ADD**

Write every missing length
on the diagram first.

★ How often does each variant come up?

22 past-paper questions (Jan 2020 → Jun 2025); 46 practice parts — grouped by type.

	PAST PAPERS (OUT OF 22)	PRACTICE PARTS
V1 Standard shapes	4	13
V2 Compound shapes	4	18
V3 Shapes with circles	6	9
V4 Working backwards	8	6

WHAT THIS TELLS YOU On every paper (Q3–Q11, 3–6 marks). The favourite is backwards: the area is given and you find a length — write the area in terms of x and solve.

BEFORE YOU START

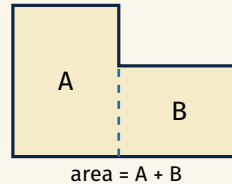
The big idea



Area is inside; perimeter is the fence.

Area (cm^2): split the shape into pieces you know and add them (or take a piece away).

Perimeter (cm): walk round the **outside** only — lines inside the shape do not count.


The formulas
Rectangle lw

Parallelogram:
 bh .

Triangle $\frac{1}{2}bh$

Or $\frac{1}{2}ab \sin C$.

Trapezium $\frac{1}{2}(a + b)h$

a, b the parallel sides.

Circle $\pi r^2, 2\pi r$

Semicircle: halve both, add the diameter for a perimeter.

Units
Hectare $1 \text{ ha} = 10\,000 \text{ m}^2$

A 100 m square.

 m^2 to cm^2 $\times 10\,000$

Square the 100.

Height $\perp$ to the base

Not the slanted side.

**CALCULATOR FIRST, THEN CHECK**

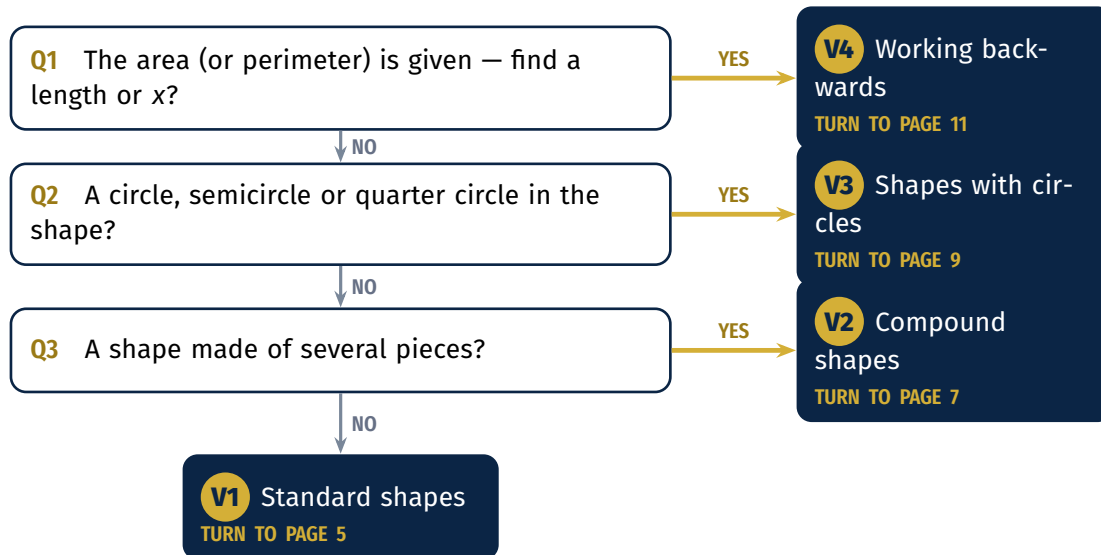
Exact answers keep π and surds: $72 - 18\pi$. "3 s.f." means calculator at the very end.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Standard Shapes

Pick the formula; find the perpendicular height (trig or Pythagoras if needed).

SEEN IN PAST PAPERS

4 / 22

PRACTICE QUESTIONS: 13

 Your notes

Came up in: Oct/Nov 2020 P1H Q11 • Nov 2021 P2H Q10 • Jun 2023 P2HR Q5 • Jun 2025 P1H Q9

? What is it?

In plain words

Name the shape, write its **formula**, then fill it in. The height is always **perpendicular** to the base.

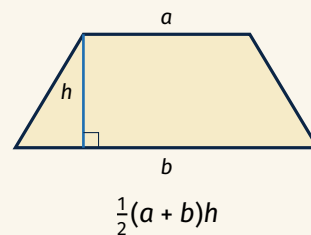
A slanted side with an angle? Use **trigonometry** for the height:
 $h = 12 \sin 60^\circ$.

IT LOOKS LIKE

trapezium field

isosceles trapezium

hectares



How do I do it?

1 Name

Rectangle, triangle, trapezium, parallelogram?

2 Height

Perpendicular; trig / Pythagoras if hidden.

3 Formula

Substitute.

4 Units cm^2 , m^2 , hectares; round at the end.

HEIGHT $\perp$ BASE Never the slanted side.

V1 Standard Shapes

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"A field is in the shape of a trapezium ... price per hectare" → **Area** $\div 10\,000$

"one line of symmetry ... angle $ADC = 60^\circ$ " → **Height** $= 12 \sin 60^\circ$; **the overhangs are equal**

"Work out the perimeter of triangle ABC" → **Missing sides by trig**

WORKED EXAMPLE **PAST PAPER**

June 2023 · Paper 2HR · Q5 · 4 marks

A field is a trapezium with parallel sides 170 m and 330 m and height 240 m. It sells for \$49 650. 1 hectare = 10 000 m². Work out the price per hectare.

STEP 1 $\frac{1}{2}(170 + 330) \times 240$ ◀ trapezium

STEP 2 $= 60\,000 \text{ m}^2$ ◀ area

STEP 3 $60\,000 \div 10\,000 = 6 \text{ ha}$ ◀ hectares

STEP 4 $49\,650 \div 6 = 8275$ ◀ price per hectare

ANSWER \$8275 per hectare $\checkmark 6 \times 8275 = 49\,650 \checkmark$ **WATCH OUT** $\times 170 \times 240$ or 330×240 $\checkmark \frac{1}{2}(170 + 330) \times 240$

Average the parallel sides first.

EXAM TIP

Write the formula before the numbers — it earns the method mark.

Check yourself

calculator first · answers on page 13

1.1 Dr Eslam

A trapezium has parallel sides 9 cm and 15 cm and height 6 cm. Area?

Answer:

1.2 Jun 2025 P1H

Isosceles trapezium: $DC = 47$, $AD = 12$, angle $ADC = 60^\circ$. Area (3 s.f.)?

Answer:

1.3 Nov 2021 P2H

$BA = BC$, $BN \perp AC$, $AN = 9.3$ cm, angle $BAC = 38^\circ$. Perimeter of ABC?

Answer:

V2

VARIANT 2

Compound Shapes

Split into standard pieces, find the missing lengths, add (or subtract) the areas.

SEEN IN PAST PAPERS

4 / 22

PRACTICE QUESTIONS: 18


 Your notes

Came up in: Jan 2021 P1HR Q10 • Jan 2021 P2H Q20 • Jun 2024 P2HR Q10 • Oct/Nov 2020 P1HR Q22

? What is it?

In plain words

Draw lines that cut the shape into **rectangles, triangles and trapeziums**.

Missing lengths come from **subtracting** the sides you know, or from **Pythagoras**.

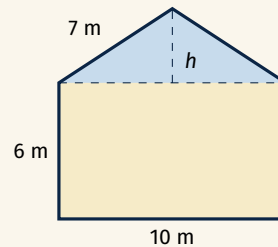
Sometimes it is easier to take a piece **away** from a big rectangle.

IT LOOKS LIKE

pentagon wall

hexagon

tins of paint



How do I do it?

1 Split

Dashed lines into standard shapes.

2 Missing lengths

Subtract, or Pythagoras / trig.

3 Areas

Each piece, then add or subtract.

4 Context

Tins, boxes, packs: divide and round up.

SPLIT, LABEL, ADD

Round up for tins.

V2 Compound Shapes

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"a pentagon with exactly one line of symmetry"

→ **Rectangle + isosceles triangle**

"least number of tins ... each covers 16 m²"

→ **Area ÷ 16, round up**

"regular octagon ... shaded region"

→ **Octagon minus the unshaded pieces**

**WORKED EXAMPLE** **PAST PAPER**

January 2021 · Paper 1HR · Q10 · 5 marks

A wall face is a pentagon: a 10 m by 6 m rectangle with an isosceles triangle of sides 7 m, 7 m on top. Each tin covers 16 m². Work out the least number of tins.

STEP 1 $h = \sqrt{7^2 - 5^2} = \sqrt{24}$ ◀ triangle height (Pythagoras)

STEP 2 $10 \times 6 + \frac{1}{2} \times 10 \times \sqrt{24}$ ◀ rectangle + triangle

STEP 3 $= 60 + 24.49 = 84.49 \text{ m}^2$ ◀ area

STEP 4 $84.49 \div 16 = 5.28 \Rightarrow 6$ ◀ round up

ANSWER 6 tins✓ 5 tins cover only 80 m² ✓**WATCH OUT**

✗ triangle height 7 m

✓ $\sqrt{24}$ m

7 m is the sloping side, not the height.

**EXAM TIP**

Write each piece's area inside it on the diagram, then add.

**Check yourself**

calculator first · answers on page 13

2.1

Dr Eslam

A 10 m by 8 m rectangle with a 4 m by 3 m corner removed. Area and perimeter?

Answer:

2.2

Jun 2024 P2HR

Hexagon: rectangle 23 by 4.7 below a trapezium with top $AB = 11$, slant $BC = 5$ at 30° to FC . Area?

Answer:

2.3

Oct/Nov 2020 P1HR

Regular octagon of side 10 cm. Find the area of $ACDEH$ (nearest cm²).

Answer:

V3

VARIANT 3

Shapes with Circles

Find the radius; full circle, then the fraction ($\frac{1}{2}$, $\frac{1}{4}$); add or subtract the straight parts.

SEEN IN PAST PAPERS

6 / 22

PRACTICE QUESTIONS: 9



Your notes

Came up in: Jan 2020 P1HR Q5 • Jan 2020 P2H Q11 • Jan 2021 P2HR Q2 • Jan 2021 P2H Q8 • Nov 2021 P1H Q4 • Jun 2022 P2HR Q5

? What is it?**In plain words**

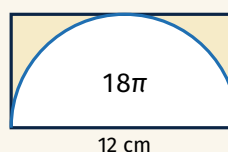
A semicircle is **half** a circle: area $\frac{1}{2}\pi r^2$, curved edge πr . Watch for **diameter given** — halve it for r .

Shaded region = big shape – the parts **not** shaded.

IT LOOKS LIKE

semicircle garden

rectangle minus semicircle



$$72 - 18\pi = 15.5 \text{ cm}^2$$

How do I do it?**1 Radius**Diameter $\div 2$ — check what is given.**2 Fraction**Whole circle, then $\times \frac{1}{2}$ or $\frac{1}{4}$.**3 Combine**

Add or subtract the other pieces.

4 Perimeter

Curved part + straight edges on the outside.

DIAMETER $\div 2$ FIRSTExact: keep π .

V3 Shapes with Circles

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"semicircle of radius 7.2 m ... enough grass seed?" → **Area, compare, then say Yes / No**

"a semicircle with diameter AB = 12 cm" → **$r = 6$**

"square and 4 identical semicircles" → **Side = $\sqrt{\text{area}}$; 4 halves = 2 circles**

**WORKED EXAMPLE** **PAST PAPER**

January 2021 · Paper 2HR · Q2 · 3 marks

A rectangle ABCD has a semicircle on diameter AB = 12 cm touching DC. Work out the area of the shaded region (rectangle outside the semicircle), to 3 s.f.

STEP 1 $r = 6$, rectangle $12 \times 6 = 72$ ◀ height = radius

STEP 2 $\frac{1}{2}\pi(6)^2 = 18\pi$ ◀ semicircle

STEP 3 $72 - 18\pi = 15.45 \dots$ ◀ subtract

STEP 4 ≈ 15.5 ◀ 3 s.f.

ANSWER 15.5 cm^2

✓ less than half the rectangle ✓

**WATCH OUT** $\times \pi \times 12^2$ $\checkmark \pi \times 6^2$

12 cm is the diameter.

**EXAM TIP**

Keep 18π until the last line — rounding early can cost the accuracy mark.

**Check yourself**

calculator first · answers on page 13

3.1

Jan 2020 P1HR

A semicircular garden has radius 7.2 m. 12 boxes each cover 6 m^2 . Enough seed?

Answer:

3.2

Jun 2022 P2HR

A square of area 36 cm^2 has a semicircle on each side. Total area (1 d.p.)?

Answer:

3.3

Dr Eslam

A 10 cm square with a semicircle on one side. Perimeter (3 s.f.)?

Answer:

V4**VARIANT 4** MOST COMMON IN THE EXAM

Working Backwards

Write the area in terms of the unknown, set it equal to the given area, solve.

SEEN IN PAST PAPERS

8 / 22

PRACTICE QUESTIONS: 6



Your notes

Came up in: Jan 2020 P1HR Q10 · Jan 2020 P1H Q3 · May 2021 P1H Q3 · Jun 2022 P1HR Q11 · Jun 2022 P2H Q5 · Nov 2023 P2H Q10

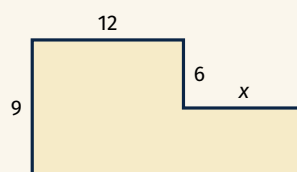
· May 2024 P1HR Q8 · Jun 2024 P2H Q7

? What is it?

In plain words

The area (or perimeter) is **given**; a length is **unknown**. Split the shape as usual, but write each piece with **x**. Add the pieces, set equal to the given total, and **solve** (Chapter 2).

IT LOOKS LIKE

area 129 cm^2 find **x**

$$9(12 + x) - 6x = 129$$

How do I do it?

1 Split

Pieces as usual.

2 In **x**

Each piece's area with the unknown.

3 Equation

Total = given area.

4 Solve

Then answer the length asked (perimeter, angle ...).

AREA IN **x = GIVEN**

The most common variant here.

V4 Working Backwards

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"The shape has area 129 cm^2 . Work out x " → **Area in $x = 129$**

"The perimeter of the hexagon equals the perimeter of the triangle" → **Two perimeters equal**

"Work out the perimeter" after finding a length → **Pythagoras for the slanted side**

WORKED EXAMPLE **PAST PAPER**

January 2020 · Paper 1H · Q3 · 4 marks

An L-shape: top side 12 cm, left side 9 cm; the top-right corner, 6 cm deep and x cm wide, is cut away. The area is 129 cm^2 . Work out x .

STEP 1 $9(12 + x)$ ◀ the full rectangle

STEP 2 $- 6x$ ◀ the missing corner

STEP 3 $108 + 9x - 6x = 129 \Rightarrow 3x = 21$ ◀ equation

STEP 4 $x = 7$ ◀ solve

ANSWER $x = 7$ $\checkmark 9 \times 12 - 6 \times 7 = 108 - 42 = 66 \checkmark$ **WATCH OUT**

$$\times 12 \times 9 + 6x = 129$$

$$\checkmark 9(12 + x) - 6x$$

Draw the pieces carefully — the corner is taken away.

EXAM TIP

Substitute your x back into the area — it should give the number in the question.

Check yourself

calculator first · answers on page 13

4.1 Jun 2024 P2H

Six-sided shape: rectangle 21×13 below a trapezium with top $CD = 15$; total height h ; area 390 cm^2 . Find h .

Answer:

4.2 May 2024 P1HR

Right trapezium: $AD = 15$, $DC = 14$, angles A and D are 90° ; area 360 cm^2 . Perimeter?

Answer:

4.3 Nov 2023 P2H

Hexagon $A(0, 0)$, $B(0, 25)$, $C(x + 2, 25)$, $D(x + 2, 17)$, $E(x + 6, 7)$, $F(x + 6, 0)$; area 258 cm^2 . Find x .

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 72 cm^2

1.2 426 cm^2

1.3 42.2 cm

2.1 $68 \text{ m}^2, 36 \text{ m}$

2.2 150.6 cm^2

2.3 327 cm^2

3.1 $81.4 \text{ m}^2 > 72$: No

3.2 92.5 cm^2

3.3 45.7 cm

4.1 $h = 19.5$

4.2 88 cm

4.3 $x = 6.4$

THE WHOLE TOPIC ON ONE PAGE

Area cheat sheet



Your notes
V1 Standard 4/22

$$\frac{1}{2}(a + b)h$$

Height $\perp$ base.**V2 Compound 4/22**

$$60 + 5\sqrt{24}$$

Split, label, add; round tins up.

V3 Circles 6/22

$$72 - 18\pi$$

Radius first; fraction of a circle.

V4 Backwards 8/22

$$108 + 3x = 129$$

Area in x, solve.

**TOP 3 LOST MARKS**

1. slanted side used as height $\times$
2. diameter used as radius $\times$
3. inside lines in a perimeter $\times$

BOTTOM LINE Split, label, add · height $\perp$ base · radius = $\frac{d}{2}$ · given area:
equation in x.